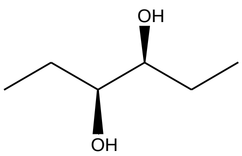
ACML 100 **MINOR** **Max. Marks: 30** **Time: 2 hours**

Name:

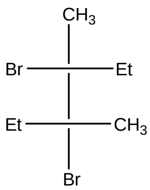
Entry number:

Q1. What is [α] for a mixture of enantiomers when the mixture is 70% R and 30% S and [α] for R is -20°? (1)

Q2. Which of the following is(are) meso compound(s)? (1)



a



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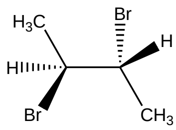
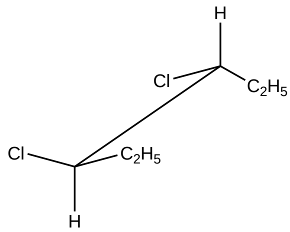
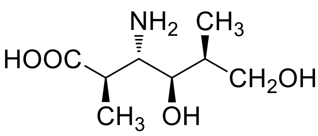


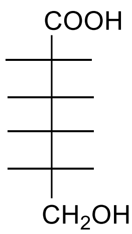
c



d

Q3. Convert the following structure into Fischer Projection in the template provided. (1)

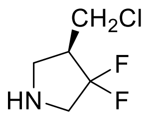


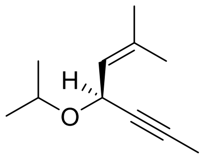


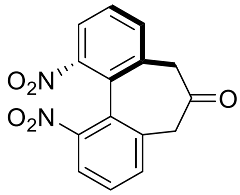
(Template)

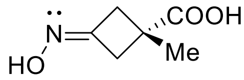
Q4. Draw the most stable Newmann projection for (2R,3R)-3-fluorobutan-2-ol. (1)

Q5. Assign absolute configuration to the compounds given below. (0.5×4=2)

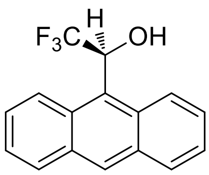
a) 

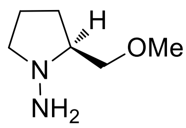
b) 

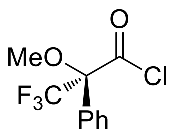
c) 

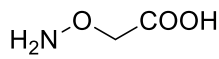
d) 

Q6. Which one of the following reagents can bring about the chiral resolution of a racemic mixture of 2-butanol. Write the reaction involved and represent schematically the procedure that will be adopted for getting enantiopure compounds. (2)









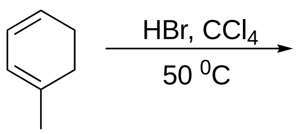
Q7. Furan on treatment with maleic anhydride at room temperature gives a compound **P**, which on heating at 120 °C changes to compound **Q**. Both **P** and **Q** have identical molecular formula. (3)

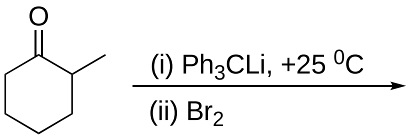
(a) Give the chemical structures of **P** and **Q**. (1)

(b) How are **P** and **Q** related to each other? (0.5)

(c) Draw the energy profile diagram for the formation of **P** and **Q** on the same graph. (1.5)

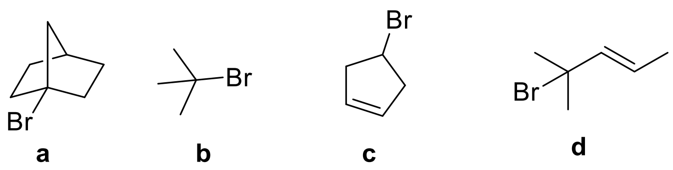
Q8. a) Write the major product for the following reactions. (1×2=2)

a) 

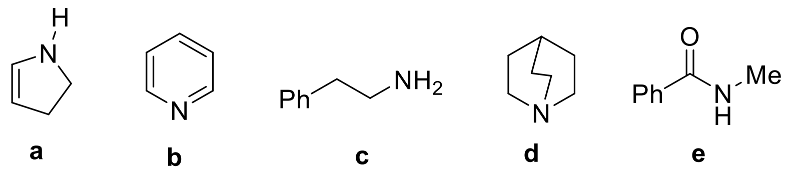
b) 

Q9. Arrange the following in ascending order of the indicated property (1×3=3)

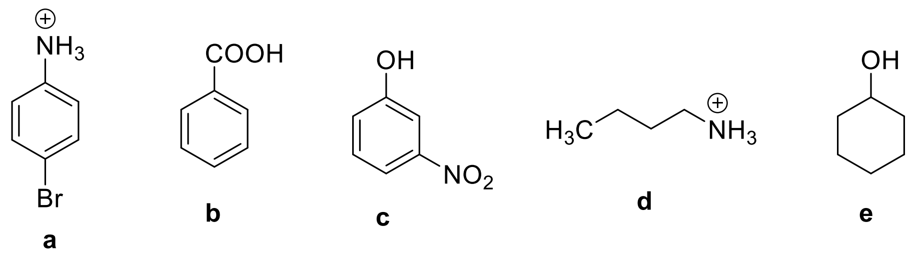
(a) Relative rate of solvolysis in EtOH-H_2O (1)



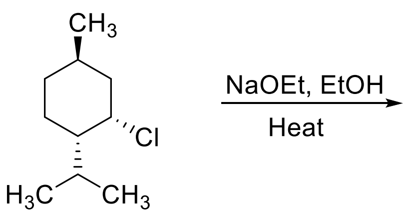
(b) Relative order of basicity (1)

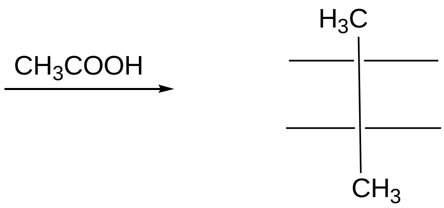


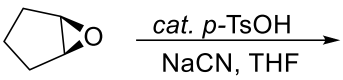
(c) Relative order of acidity (1)



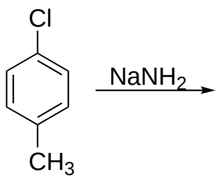
Q10. Give the structure of the product(s) for the following reactions with correct stereochemistry wherever applicable. (1×3=3)

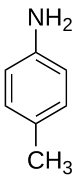
a) 

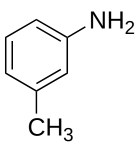
b) (*erythro*)-3-phenyl-2-butyl tosylate 

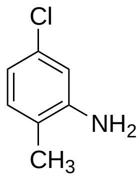
c) 

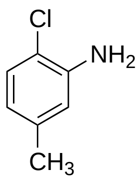
Q11. Which is the major product of the following reactions. (1×3=3)

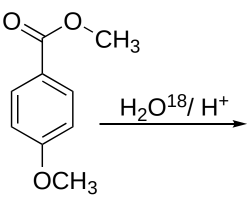
i) 

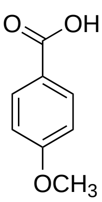
a) 

b) 

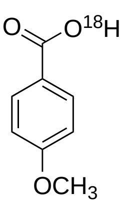
c) 

d) 

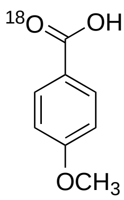
ii) 

a) 

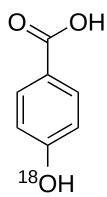
+ $CH_3O^{18}H$

b) 

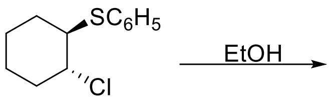
+ CH_3OH

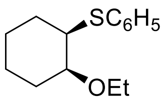
c) 

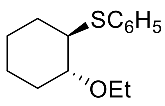
+ CH_3OH

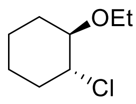
d) 

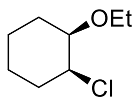
+ CH_3OH

iii) 

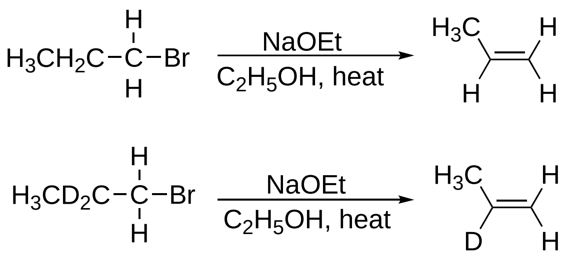
a) 

b) 

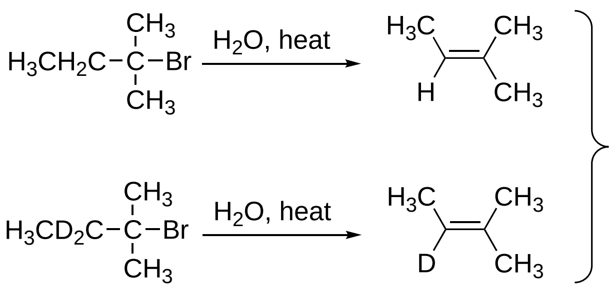
c) 

d) 

Q12. Explain the following observations with suitable explanation showing any transition state/intermediate involved. (2)

A) 

$K_H / K_D = 6.7$

B) 

$K_H / K_D = 1.4$

Q13. A sample (5.6 g) containing iron is completely dissolved in cold dilute HCl to prepare a 250 mL of solution (All Fe is converted to $FeCl_2$). Titration of 25.0 mL of this solution requires 12.5 mL of 0.03 M $KMnO_4$ solution to reach the end point. The reaction of $FeCl_2$ with $KMnO_4$ is given as

$$5FeCl_2 + KMnO_4 + 8HCl \rightarrow MnCl_2 + 5FeCl_3 + 4H_2O + KCl$$

Calculate the number of moles of Fe present in 250 mL solution. (3)

Q14. Sulphur trioxide, a precursor of sulfuric acid, is prepared industrially using the contact process by passing sulphur dioxide and oxygen over a hot catalyst:

$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons SO_{3\,(g)}$$

Under standard condition, ΔH = -196 kJ mol^{-1}, ΔS = -188 J K^{-1}mol^{-1} for the process.

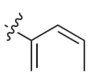
(i) Does an increase in temperature favour an increase in product at equilibrium ? (1)
(ii) Does an increase in pressure favour an increase in product at equilibrium ? (1)
(iii) Is the reaction spontaneous at 800 K ? (1)

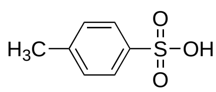
(Assume the ΔH and ΔS do not change with temperature)

Abbreviations:

-Me = $-CH_3$

-Et = $-CH_2CH_3$

-Ph = 

p-TsOH = 

THF=Tetrahydrofuran